UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

FACULTAD DE INGENIERÍA DE SISTEMAS E INFORMÁTICA

DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING TECHNOLOGIES



RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED TECHNOLOGIES

Deliverable:

Model Card and Data Sheet

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1. Purpose and Research Status

This document consolidates the two documentation artifacts required in Session 7: a **Model Card** describing the planned machine-learning artifact, and a **Dataset Datasheet** describing the planned research dataset. It is aligned with the course requirement to document intended use, limitations, subgroup considerations, data provenance, preprocessing, misuse risks, and reproducibility.

> **Research-stage honesty rule**
>
> The final model and individual-level dataset have not yet been fixed or trained. Therefore, this document does not invent performance values, thresholds, licenses, collection details, or subgroup results. Fields that remain unresolved are explicitly marked "UNKNOWN — to investigate". This is an intentional methodological choice, not an omission.

Documentation objectives

- Document the planned model using the 9 Model Card sections introduced by Mitchell et al. (2019).
- Document the planned dataset using the 7 lifecycle sections of Datasheets for Datasets introduced by Gebru et al. (2021).
- Make intended use and misuse explicit, especially because the artifact concerns educational risk and potentially sensitive student records.
- Prepare the research workflow for reproducible evaluation, subgroup analysis, leakage control, multiple random seeds, and future external validation.

2. Model Card — XAI School Interruption Risk Model

1. Model details

Field	Current research documentation
Name / version	XAI School Interruption Risk Model — Research Prototype v0.1
Owner / contact	Cendik — Doctoral Program in Deep Technologies, UNMSM
Date	October 2026
Architecture	Candidate supervised-learning pipeline. Final algorithm is not yet fixed; planned comparison includes Logistic Regression, Decision Tree, Random Forest, XGBoost/LightGBM, and potentially a neural network. Explainability is planned through SHAP and/or LIME.
License	UNKNOWN — to investigate before model release.

2. Intended use

Field	Current research documentation
Primary intended uses	Research prototype to support early identification of secondary-education students in Peru who may be at risk of school interruption, so that authorized educational professionals can prioritize preventive support.
Out-of-scope / misuse	NOT for automatic exclusion, punishment, denial of educational services, disciplinary action, or fully automated decisions about a student. NOT a substitute for professional educational judgement or direct contact with the student and family.
Intended users	Researchers and, only after validation and governance approval, authorized educational professionals and decision-makers involved in student-support interventions.

3. Factors

Field	Current research documentation
Relevant groups / conditions	Potential factors include academic, demographic, socioeducational and territorial conditions. The final subgroup definitions depend on variables legitimately available in the selected MINEDU or other authorized educational dataset.
Evaluation factors	Planned disaggregation by relevant subgroups and conditions such as geography, school context and other legally/ethically appropriate variables available in the final dataset. Exact subgroup list: UNKNOWN — to investigate after dataset finalization.

4. Metrics

Field	Current research documentation
Performance measures	Planned: Precision, Recall, F1-score, ROC-AUC and confusion matrix. Accuracy may be reported but will not be used alone because school interruption may be an imbalanced outcome.
Decision thresholds	UNKNOWN — to investigate. Thresholds must be chosen on validation data according to the cost of false negatives and false positives.
Variation approach	Planned: repeated evaluation over 5+ random seeds and reporting mean ± standard deviation; confidence intervals will be considered for principal performance estimates.

5. Evaluation data

Field	Current research documentation
Datasets	A held-out test partition from the finalized Peruvian secondary-education dataset, with temporal and/or territorial external validation if the data structure permits.
Preprocessing	Planned leakage-safe pipeline: split first; fit imputation, encoding, scaling, feature selection and any class-imbalance treatment on training data only. Exact procedure: UNKNOWN — to investigate after the dataset is finalized.

6. Training data

Field	Current research documentation
Datasets	Planned use of public or formally authorized Peruvian education data, with MINEDU sources as the primary candidate. The final unit of analysis and dataset version are not yet fixed.
Known limitations	Current project design does not yet establish a final individual-level longitudinal dataset. Public sources may be aggregated, incomplete, non-longitudinal, or insufficiently representative for individual risk prediction.

7. Quantitative analyses

Field	Current research documentation
Overall results	UNKNOWN — to investigate. The final model has not yet been trained and no performance number is claimed.
Disaggregated results	UNKNOWN — to investigate. Planned reporting by relevant subgroups/conditions after the final dataset and protected/ethical analysis variables are established.

8. Ethical considerations

Field	Current research documentation
Risks	False positives may stigmatize students; false negatives may miss students who need support. Historical inequities may be learned from the data. Explanations can be misinterpreted as causal. Educational records may contain sensitive personal information.
Mitigations	Human-in-the-loop use; fairness analysis; privacy protection and data minimization; transparent explanations; restricted access; monitoring for disparate performance; no punitive automated decisions; explanations framed as associations rather than causal proof.

9. Caveats and recommendations

Field	Current research documentation
Caveats	This is currently a research prototype, not a deployed system. Dataset, final model, threshold, external validation, subgroup performance and operational governance remain to be established.
Recommendations	Do not deploy until predictive validity, explanation faithfulness/stability, subgroup fairness, privacy, temporal/territorial robustness and human oversight have been evaluated. Reassess performance periodically because educational conditions may drift over time.

3. Dataset Datasheet — Planned Peruvian Secondary-Education Dataset

1. Motivation

Field	Current research documentation
Purpose	To support doctoral research on an Explainable Artificial Intelligence artifact for early identification of school interruption risk among secondary-education students in Peru.
Created by / funded by	Dataset to be assembled from public or formally authorized educational sources, with MINEDU as the primary candidate source. Exact dataset creator, custodian and funding metadata: UNKNOWN — to investigate.

2. Composition

Field	Current research documentation
What instances represent	Intended unit: records describing secondary-education students or the closest ethically and legally usable unit available in the selected source. Final unit of analysis: UNKNOWN — to investigate.
Labels	Intended target: school continuity/interruption risk. Exact operational definition, observation window and labeling rule: UNKNOWN — to investigate.
Sample vs population	Target population: secondary-education students in Peru. The eventual dataset will be a sample or administrative subset and must not be assumed to represent every student equally.
Missing data	UNKNOWN — to investigate after the final dataset is selected. Missingness patterns must be documented and assessed for systematic bias.

3. Collection process

Field	Current research documentation
How collected	Expected to originate from administrative or open educational records published or authorized by Peruvian education authorities. Exact collection instruments, responsible entities and linkage process: UNKNOWN — to investigate.
Timeframe	UNKNOWN — to investigate. The final study should document school years covered and whether data are cross-sectional or longitudinal.
Consent / ethics	UNKNOWN — to investigate for the final source. Public/administrative availability does not by itself remove privacy, governance or ethical obligations.

4. Preprocessing / cleaning / labeling

Field	Current research documentation
Steps	Planned: data-quality checks, duplicate handling, missing-value analysis, encoding of categorical variables, leakage review, class-imbalance assessment and reproducible train/validation/test partitioning. All learned preprocessing must be fitted on training data only.
Raw data available	UNKNOWN — to investigate. Availability will depend on the final MINEDU or other authorized source and applicable access/privacy restrictions.

5. Uses

Field	Current research documentation
Appropriate uses	Research on early-warning modeling, XAI, methodological evaluation, fairness analysis and decision-support design for preventive educational interventions.
Should NOT be used for	Should NOT be used for punitive decisions, exclusion from services, student ranking, disciplinary sanctions, deterministic labeling of a student as a dropout, or any fully automated high-impact educational decision.

6. Distribution

Field	Current research documentation
How distributed	UNKNOWN — to investigate. Distribution will follow the terms of the final data source; sensitive or restricted student-level data must not be publicly redistributed.
License / restrictions	UNKNOWN — to investigate for each source used. Access, redistribution and derivative-use conditions must be recorded before release.

7. Maintenance

Field	Current research documentation
Maintainer	Primary source custodian: UNKNOWN — to investigate. Research-derived dataset documentation would be maintained by the doctoral researcher during the study.
Update plan	UNKNOWN — to investigate. If the artifact is used beyond research, periodic updates and drift monitoring would be required as educational conditions and administrative definitions change.

4. Documentation Self-Audit

The course rubric emphasizes **completeness**, **honesty**, **disaggregation**, **reproducibility**. The current documentation is evaluated below. Passing a documentation check does not mean the underlying model is already scientifically validated; it means the required information is explicitly documented or the gap is honestly flagged.

Documentation check	Status	Rationale
Model Card has all 9 sections	**PASS**	All nine sections are present.
Datasheet has all 7 sections	**PASS**	All seven lifecycle sections are present.
Intended misuse / out-of-scope stated	**PASS**	Automated punitive or exclusionary use is explicitly prohibited.
Performance variation planned	**PASS**	Evaluation over 5+ seeds with mean ± standard deviation is planned; final measured values remain pending.
Metrics planned for disaggregation	**PASS**	Subgroup/condition reporting is explicitly planned, while the exact subgroup list remains pending.

Dataset “should NOT be used for” stated	**PASS**	High-impact punitive and fully automated educational decisions are explicitly excluded.
Collection process addressed	**PASS**	Expected source type is described; unresolved collection details are explicitly flagged.
Unknowns flagged honestly	**PASS**	Unresolved methodological details are marked “UNKNOWN — to investigate”.

Documentation score: 8/8 checks passed

Interpretation of the 8/8 score

This score reflects documentation completeness according to the course-style checks. It does NOT claim that the research model has achieved 100% scientific validity, reproducibility, fairness, or deployment readiness. Several substantive research components remain explicitly unresolved and must be investigated before empirical claims are made.

5. Priority Items to Resolve Before the Empirical Phase

1. Finalize the data source: Identify the exact MINEDU or other authorized dataset, version, custodian, access conditions, temporal coverage, unit of analysis and redistribution restrictions.

2. Operationalize the target: Define precisely what counts as school interruption, the prediction horizon, observation window and labeling rule.

3. Define the benchmark protocol: Establish train/validation/test splitting, baseline models, hyperparameter search, multiple seeds, leakage controls and the statistical comparison plan.

4. Define subgroup and fairness analysis: Determine which subgroup variables can be legally and ethically analyzed, then predefine fairness and disaggregated performance checks.

5. Validate explanations: Evaluate not only predictive performance but also the faithfulness, stability and usefulness of SHAP/LIME explanations for intended educational users.

6. Define governance and human oversight: Specify who may access predictions, how interventions are triggered, how students can be protected from stigmatization, and how model decisions will be monitored and reviewed.

6. Session 7 Documentation References

Mitchell, M. et al. (2019). Model Cards for Model Reporting. Proceedings of FAccT 2019.

Gebru, T. et al. (2021). Datasheets for Datasets. Communications of the ACM, 64(12), 86–92.